

AI Usage Note — NIFTY Event-Driven Recovery Study

Purpose: AI supported my research and implementation, but I remained responsible for the methodology, research decisions, interpretation and final conclusion.

1. AI Tools and My Working Process

ChatGPT: I mainly used it to understand concepts, compare methodological alternatives and challenge assumptions around look-ahead bias, overlapping observations, baseline contamination, statistical testing, robustness, multiple testing and interpretation. **Claude Code:** I mainly used Claude inside VS Code for implementation, debugging, experiments, notebook structure and reproducing results. My general workflow was: **understand the decision** → **consider alternatives** → **make my decision** → **ask Claude to implement** → **inspect the output** → **challenge the result**.

2. Main Decisions — AI Input vs My Final Decision

Decision	AI discussion / alternatives	My final decision and reasoning
Significant fall	Discussed fixed-percentage rules and volatility-scaled rules, including 1.5σ , 2σ , 2.5σ and different lookback windows.	I chose -2σ using the previous 60 trading days . I wanted an unusually large fall relative to recent volatility rather than an arbitrary fixed percentage. Other thresholds became robustness checks.
Recovery	Discussed different recovery levels and ways to define a rebound.	I chose 50% recovery within 5 trading days . Example: 20,000 → 19,000 means 19,500 is the recovery level. I kept this fixed instead of choosing the definition with the strongest result.
Holding period	AI helped compare 1, 3, 5 and 10 days and highlighted post-event selection risk.	I chose 5 days as the main period and used 1/3/10 as robustness. I rejected 'wait for the first positive day' because the exit would depend on observing the future path.
Entry / exit	Discussed event-close versus next-day execution and look-ahead bias.	I chose next-day Open → day-5 Close because the event is only known after the event-day close. Next-day Close was a robustness check.
Overlapping events	AI explained dependence from overlapping 5-day windows and discussed episode grouping and block bootstrap.	I kept event-level analysis but grouped nearby events into episodes. I also tested different block lengths and two backtest rules because overlap could materially change the result.
Statistics	AI explained why independent-observation tests can be inappropriate with overlapping returns and suggested block bootstrap. Newey-West was added as a secondary check.	I used block bootstrap as the main method: 2,000 repetitions, 95% CI and $L=10$. I added Newey-West as a secondary check and obtained the same practical conclusion for returns.
OOS split	AI emphasized chronological testing to avoid future information influencing development decisions.	I used 2000–2017 development / 2018–2026 OOS . I wanted the setup frozen before evaluating the later period.
Costs	Discussed realistic execution and transaction-cost sensitivity.	I chose 0.10% round-trip cost and tested 0.05%/0.20%. The cost was an assumption, not a claim about the historical average.

3. My Own Research Considerations

- **I did not want to optimize for the best-looking number.** When several thresholds and holding periods were available, I decided that the main specification should not be selected because it produced the highest return. I kept the 2σ / 60-day / 5-day setup as the main specification and treated alternatives as robustness checks.
- **I questioned the 'first positive day' idea.** I considered whether waiting for the first positive day after a fall would be a better exit rule, but I realized that the future path would then determine the exit. I therefore did not use it as the main strategy because it could introduce post-event selection.
- **I wanted to challenge the positive backtest.** The non-overlapping backtest looked positive OOS, but I did not stop there. I kept an every-event version as a challenge. Its OOS result was very different, showing that the apparent trading performance was sensitive to overlap handling.
- **I checked whether a few extreme observations were driving the conclusion.** I examined the largest events, including the COVID-period falls, and also tested the result after removing the five largest events. I did not use these checks to make the result look better; I used them to see whether the conclusion would break.
- **I investigated the pre-2011 Open-price issue.** I noticed that some older Open prices had suspicious patterns compared with later data. I checked the affected observations and ran a clean-period robustness check instead of silently ignoring the issue.
- **I treated multiple testing as a real problem.** One OOS variant produced a significant result, but it was one of many variants tested. I did not replace the main specification with that result because a significant result can occur by chance when many alternatives are tested.

4. Where I Agreed, Modified and Challenged AI

AGREED — I followed AI reasoning when it solved a clear methodological problem and matched the research objective. Examples were the chronological OOS split, using the first observable price after the signal, using a block bootstrap for overlapping observations, and testing alternative thresholds, horizons and costs. I still checked the reasoning and outputs before accepting them.

MODIFIED — AI often presented several reasonable options rather than one answer. I compared them with the actual question I was trying to answer. For example, I did not turn every tested threshold or holding period into a new main strategy. I kept one main specification and used the others for robustness. I also combined event-level analysis with episode grouping because they answer different questions about the same events.

CHALLENGED — I did not assume that a positive backtest meant the hypothesis was proven. I deliberately compared the non-overlapping and every-event backtests, checked extreme events and data-quality concerns, and separated the recovery probability from the tradable return. This led me to a more cautious final conclusion: the recovery pattern was more consistent than the return edge.

5. Coding, Writing Assistance and What I Learned

Claude Code — implementation: Claude helped turn the chosen methodology into reusable Python code, debug problems, run experiments, build the notebook sections and reproduce the headline numbers. I reviewed the outputs and did not treat generated code as automatically correct.

AI-assisted writing: I used AI **partially for both the Research Note and this AI Usage Note**. I wrote the underlying research content, decisions, results and interpretation. AI was used for limited help with **grammar correction, sentence clarity, organization, shortening repetitive wording and presentation**. Neither document was generated completely by AI as a replacement for my research reasoning. For this AI Usage Note, the decision history and description of my AI use are based on my actual project work; AI mainly helped me present it clearly.

What I learned: The biggest lesson was that finding a recovery pattern is not the same as finding a tradable return edge. The research became stronger when I treated my own result as something to challenge through OOS testing, costs, overlap assumptions, alternative definitions, extreme-event checks and multiple-testing concerns.